Kathmandu University
Department of STEAM Education
Hattiban, Lalitpur

Lead Management System (LMS)

Code No: ITEX 340

ITEX 340 Professional IT Project I

A Project Report Submitted in Partial Fulfillment of the Requirements
for Third Year/Semester I in BTech Ed IT

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Bona Fide Certificate

This project work on “Lead Management System” is the bona fide work of Santosh Bhatta, Dipshikha Ghimire and Samir Tamang who carried out the project work under my supervision.

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Date: _____

Acknowledgement

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We would also like to extend our heartfelt thanks to the Department of STEAM Education and Kathmandu University for providing us with the opportunity, resources, and academic environment necessary to carry out this project. Finally, we are grateful to all the individuals, friends, and well-wishers who directly or indirectly supported us during the completion of this project. Their encouragement and assistance played an important role in the successful completion of this work.

Abstract

Managing student leads and scholarship inquiries at Kathmandu University currently involves manual tracking through spreadsheets, emails, and disconnected systems. This creates inefficiencies and data redundancy. The purpose of this project is to design and develop a centralized Lead Management System (LMS) for Kathmandu University that automates the tracking, evaluation, and conversion of potential student leads. The system will include modules for Lead Capture and Intake, Scholarship Eligibility Checking, Course Information & Education, Follow-up & Nurturing, Application Tracking, Orientation, and User Management, designed logically and implemented using HTML, CSS & JS(frontend), Django (backend), and MySQL. The Lead Management System (LMS) will streamline admissions and outreach processes, improve lead conversion rates, and enhance transparency in scholarship eligibility and enrollment workflows. The system will contribute to Kathmandu University's digital transformation by offering real-time data access, performance metrics, and a seamless communication channel between departments and applicants. It is recommended that the system be implemented in phases, starting with core functionalities to gather early user feedback and ensure continuous improvement.

Keywords: *Learning Management System, Lead Management System (LMS), Orientation, User Management, Digital Transformation*

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Acronyms/Abbreviations

The following abbreviations are used throughout this report. Each abbreviation is also defined at first use in the text.

LMS	Lead Management System
ERD	Entity Relationship Diagram
DFD	Data Flow Diagram
HOD	Head of Department
UI	User Interface
HTML	HyperText Markup Language
CSS	Cascading Style Sheets
JS	JavaScript
RAM	Random Access Memory

Introduction

Background

Universities today face increasing challenges in managing large volumes of admission inquiries, scholarship applications, and student follow-ups. As higher education institutions continue to attract diverse applicants from different regions, the process of handling inquiries, evaluating eligibility, and ensuring timely communication becomes complex and time-consuming. Traditionally, Kathmandu University relies on manual methods such as spreadsheets, email threads, and phone records to track potential students and manage communication between departments. These manual approaches are prone to data loss, redundancy, and human error, often resulting in inconsistent communication, delayed responses, and poor lead conversion rates.

In the current context, where students expect quick responses and a seamless digital experience, manual systems fail to meet the efficiency standards required for modern academic operations. The absence of a centralized database and automated workflow limits the ability of staff and faculty members to monitor leads, evaluate scholarship eligibility, and track the student journey effectively. This leads to unnecessary administrative workload and reduces the ability of the university to make data-driven decisions regarding admissions and student engagement.

Recent advancements in educational technology emphasize the importance of adopting centralized digital systems that can automate administrative workflows. A Lead Management System (LMS) in this context serves as an integrated platform to capture, organize, and analyze every stage of a student's admission process from the initial inquiry to final enrollment. Such a system supports multiple functions, including lead capture, scholarship eligibility checks, application processing, enrollment confirmation, and orientation scheduling, ensuring a smooth and transparent admission process for both students and administrative staff.

The proposed LMS will be designed as a web-based system, accessible to authorized users from any location. It will allow administrative staff, faculty members, and department heads to manage and update lead information, monitor performance metrics, and track scholarship and enrollment status in real time. Additionally, it will provide valuable insights

into the efficiency of recruitment efforts, enabling better planning and coordination between departments. By automating repetitive tasks and centralizing communication, the system aims to enhance the overall productivity and digital transformation efforts of Kathmandu University, aligning with global trends in higher education management.

Problem Statement

The current admission and scholarship management process at Kathmandu University relies largely on manual and semi-digital methods such as spreadsheets, phone logs, and emails. These traditional approaches lead to issues like data duplication, loss of information, delayed responses, and inconsistent follow-up with potential students. As the number of inquiries grows each year, manual handling has become increasingly inefficient and prone to errors, resulting in missed enrollment opportunities and poor scholarship tracking. Another challenge is the absence of a centralized system to manage leads across departments. Each faculty maintains its own records, causing data fragmentation and making it difficult to monitor the complete admission pipeline. This lack of integration also limits the ability to assess lead quality, track communication, and evaluate scholarship eligibility in real time. Therefore, there is a clear need for a Lead Management System (LMS) that can centralize lead data, automate scholarship checks, timely follow-ups, and provide analytical insights to enhance admission efficiency and decision-making.

Objectives

The main objective of this project is to develop a Lead Management System that simplifies admission and scholarship management processes.
Specific objectives include:

- To develop a web-based system for capturing and managing student leads.
- To implement a scholarship eligibility checking module for applicants.
- To create a workflow to track student progress from inquiry to enrollment.
- To develop a dashboard displaying lead statistics and conversion metrics.

Motivation and Significance

The motivation for this project arises from the increasing complexity of managing student admissions and scholarships manually at Kathmandu University. With the growing

number of inquiries, the administrative workload has significantly increased, making it difficult to maintain timely communication and accurate data. This project aims to address those issues by creating a small, scalable digital solution that simplifies lead tracking and enhances operational efficiency.

The significance of this system lies in its potential to improve coordination between different departments, reduce repetitive manual tasks, and enhance decision-making through better data visibility. By automating scholarship eligibility checks and follow-ups, the university can ensure fairness, transparency, and faster processing. Moreover, the system supports Kathmandu University's vision of adopting modern technology to promote efficiency and digital transformation in educational management.

Related Works

Lead management systems have grown essential in higher education to automate and optimize student recruitment and admissions. Srimathi and Krishnamoorthy (2019) emphasize that effective lead management improves promotion performance and lead conversion by automating candidate engagement across multiple channels and utilizing analytics to centralize lead capture and follow-up. Their findings directly support the need for Kathmandu University's Lead Management System (LMS) to reduce redundancy and increase enrollment efficiency.

Phuengrod et al. (2021) explore integrating intelligent conversational agents within student relationship management systems. Their study highlights enhanced communication efficiency, reduction of manual workload, and timely follow-up responses to prospective students. This aligns with the project's Follow-up and Nurturing module, aiming to streamline applicant engagement and improve conversion rates. Regarding scholarship management, Olanrewaju et al. (2025) review systems incorporating biometric validations to bolster security and transparency. Their insights underscore the importance of safeguarding scholarship processes against fraud and ensuring fair, efficient application handling, which informs the design of the Scholarship Eligibility Checking module in this LMS.

Technologically, the combination of HTML, CSS, and JS for frontend, Django for backend, and MySQL for the database has been documented as a practical, scalable, and maintainable stack in various lead and learning management implementations. Projects such as Nakolah's Lead Manager Django app have demonstrated effective user management, lead tracking, and security features using these technologies, showing that the stack can handle complex workflows needed in higher education admissions (Nakolah, 2020).

Design and Implementation

The development of the Lead Management System follows an Agile methodology approach. It emphasizes incremental development, where the project is divided into smaller, manageable iterations called sprints. Each sprint delivers a functional part of the system that can be tested, reviewed, and refined before the next sprint begins.

Table 1

Development Phases

Phase	Description	Duration
Requirement Analysis	Gathering and documenting functional and non-functional requirements	Week 1-2
System Design	Creating architectural diagrams and database schema	Week 3
Database Implementation	Setting up MySQL database and tables	Week 4
Frontend Development	Building user interface with HTML, CSS, and JavaScript	Week 5-6
Backend Development	Implementing Django server-side logic and APIs	Week 7-8
Integration and Testing	Combining components and performing system testing	Week 9
Deployment and Documentation	Hosting the system and preparing technical documents	Week 10

The system architecture consists of three main components:

- Frontend Interface developed using HTML, CSS, and JavaScript
- Backend Server developed using Django.
- Database Management System using MySQL

Table 2

System Requirements

Requirement	Description	Priority
User Login	Secure authentication using username/password and session management.	High
Lead Capture	System allows capturing student inquiry details from different sources.	High
Lead Management	Admin or faculty can view, update, or delete lead information.	High
Scholarship Eligibility Check	System automatically evaluates whether a student qualifies for scholarship based on predefined criteria.	High
Lead Tracking Workflow	Track lead progress through stages such as Captured → Qualified → Applied → Enrolled → Started.	High
Dashboard Analytics	Dashboard displays statistics such as number of leads, conversion rates, and performance metrics.	Medium
Profile Management	Users can update their personal information and account details.	Medium
Communication & Follow-up	Staff can manage follow-ups with potential students and update communication status.	Medium
Report Generation	Export reports such as lead lists, scholarship status, and enrollment statistics.	Low
User Management	Multiple users, such as admin, staff, and leads, with different roles.	High

Table 3

Non-functional Requirements

Requirement	Description	Priority
Security	Password encryption and SQL injection prevention	High
Performance	Page load time under 3 seconds	High
Usability	Intuitive interface requiring minimal training	Medium
Reliability	99% uptime availability (after hosting)	High
Scalability	Support for 1000+ concurrent users	Medium
Compatibility	Cross-browser support (Chrome, Brave, Safari)	Medium

Figure 1

ER Diagram of Lead Management System (LMS)

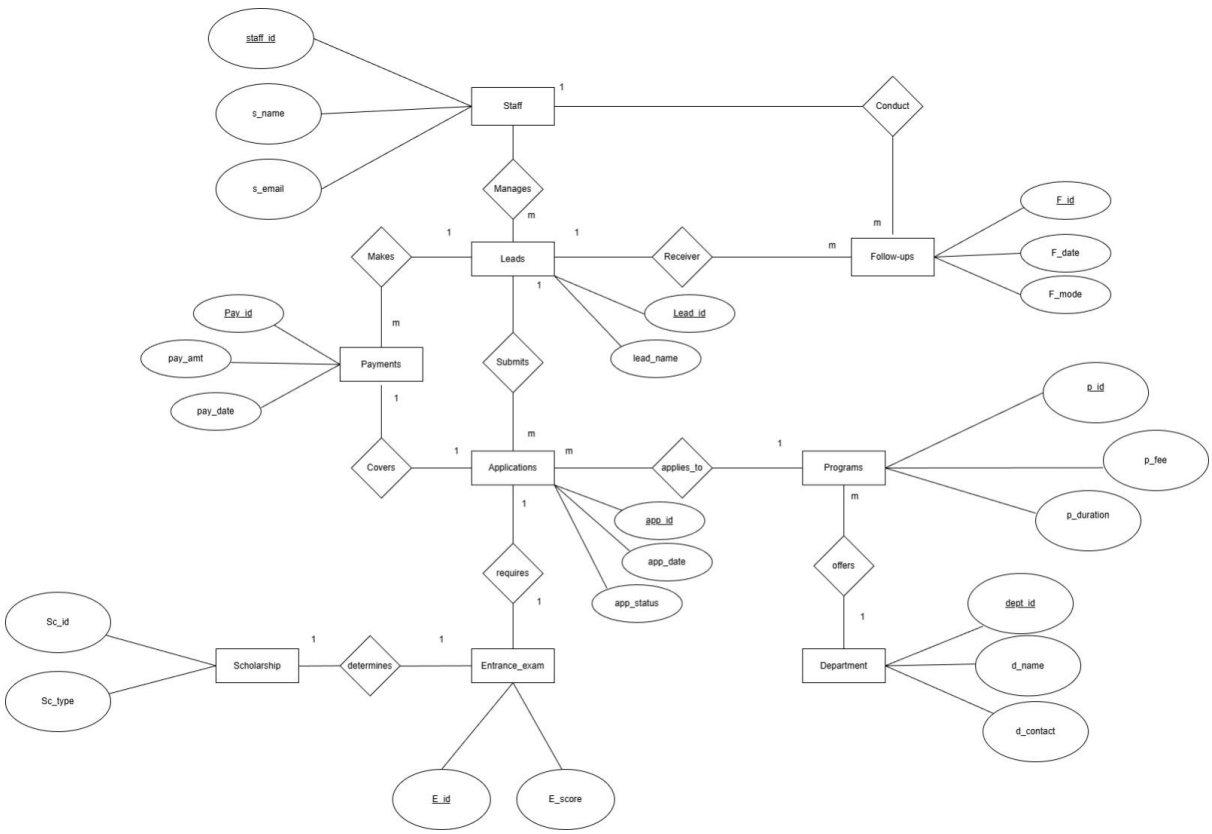


Figure 2

Flowchart of the Lead Management System (LMS)

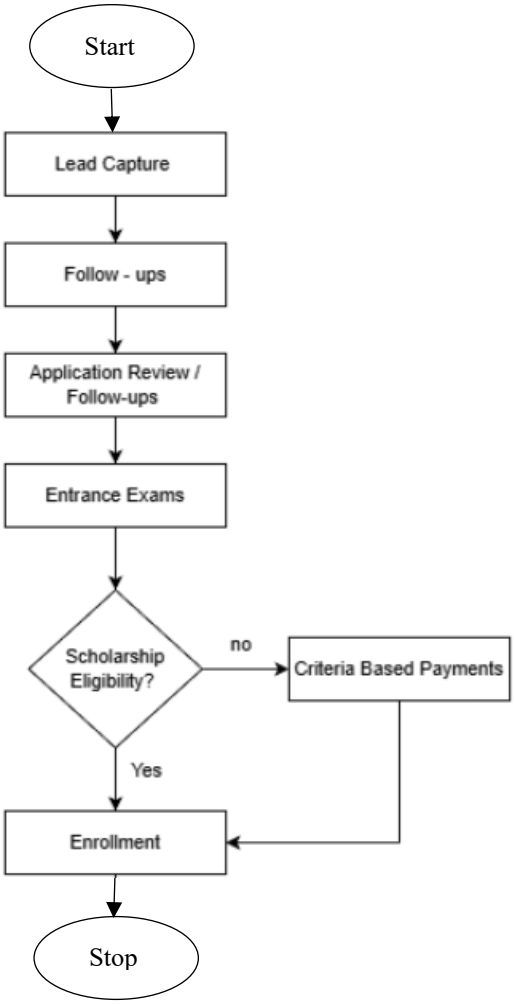
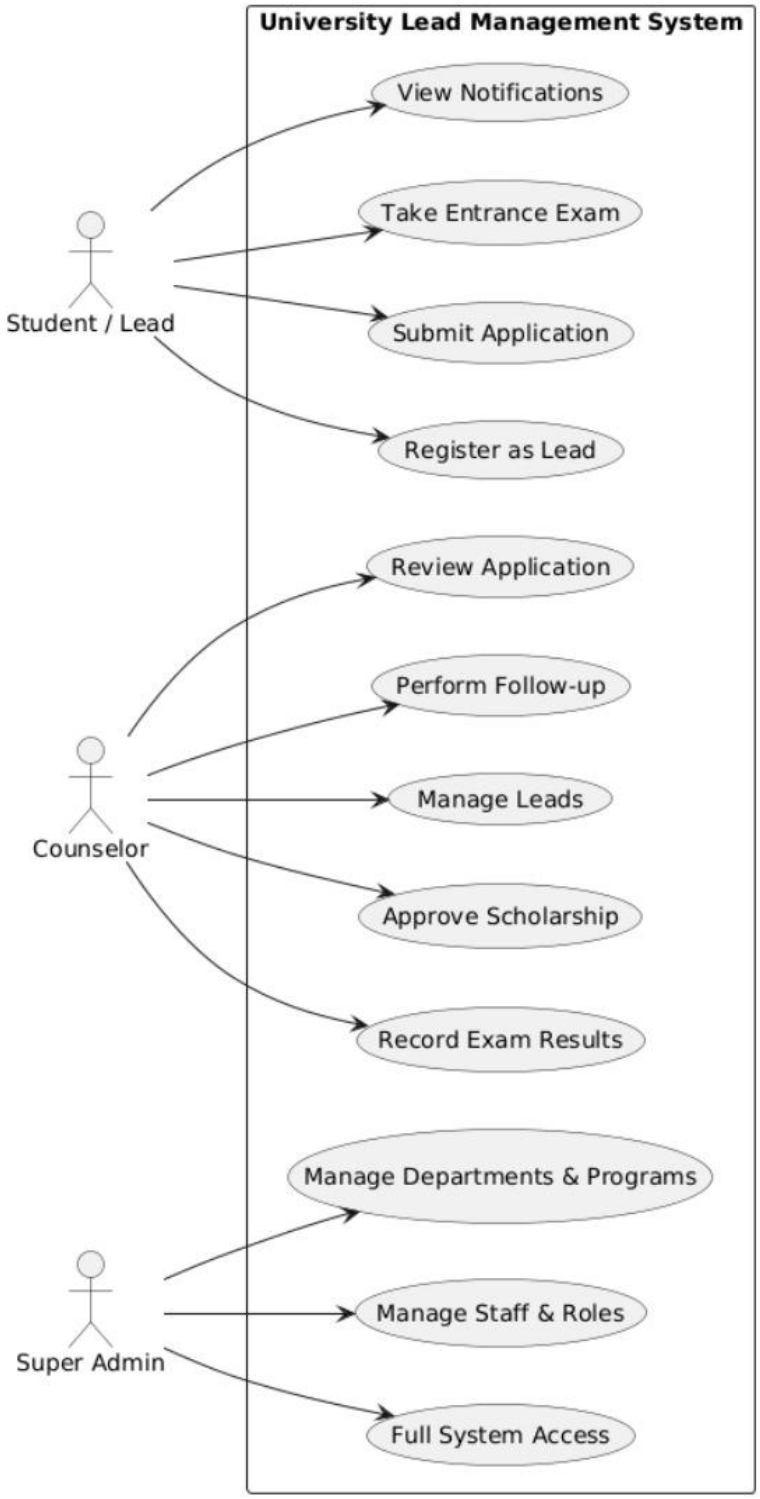


Figure 3

Physical ER of Lead Management System (LMS)

Figure 4

Use Case Diagram of Lead Management System (LMS)



System Requirement Specifications

Functional Requirements

i. Capture, view, update, and delete leads.
ii. Automatically evaluate scholarship criteria for each lead.
iii. Track payments, voucher uploads, and enrollment status.
iv. Secure login for Admin, Faculty, and HOD roles.
v. Display metrics, charts, and lead performance analysis.
vi. Provide information about schedules, canteen, and library access.

Non-Functional Requirements

i. User-friendly interface with simple navigation and consistent design.
ii. System should handle multiple user requests efficiently.
iii. Designed to accommodate future module additions or increased leads.
iv. Database backup and recovery support.
v. Accessible via major web browsers (Chrome, Edge, Firefox).

Software Specifications
Table 4

Software Specifications

S.N.	Software / Tool	Version (Example)
1	Operating System	Windows 10 / Linux / macOS
2	Backend Framework	Django 5.2
3	Frontend Framework	HTML, CSS, and JS
4	Markup Language	HTML5
5	Styling Language	CSS3
6	Scripting Language	JavaScript
7	Database Management System	MySQL 8.4
8	Code Editor / IDE	Visual Studio Code
9	Version Control System	Git / Github
10	Designing Tools	Figma, Canva, and Replit

Hardware Specifications
Table 5

Hardware Specifications

S.N.	Component	Minimum Requirement	Recommended Requirement
1	Processor	Intel Core i3 or equivalent	Intel Core i5 or higher
2	RAM	4 GB	8 GB or higher
3	Storage	20 GB free space	50 GB or more
4	Display	1366 × 768 resolution	Full HD (1920 × 1080)
5	Internet Connection	Stable broadband connection	High-speed internet

Discussion on Achievements

The development of the Lead Management System successfully demonstrates the feasibility of implementing a centralized platform for managing admission inquiries and scholarship applications.

During the project, the team designed the database structure, developed the frontend interface, and implemented backend functionalities for capturing and tracking student leads. The system also includes modules for scholarship eligibility checking and user management. Through testing and integration, the system achieved the primary objectives defined at the beginning of the project. Although certain challenges such as limited testing environments and time constraints were encountered, the system successfully demonstrates how digital technology can improve admission management processes.

Features

Features 1: "Approve, reject, and delete applications"

Features 2: "Capture, view, update, and delete leads"

Features 3: Evaluate applicants based on predefined criteria

Features 4: Visual data of lead source and lead status

Features 5: "Secure login for Admin, Leads, and Staff."

Features 6: "Metrics, charts, conversion rates, performance reports"

Features 7: Automated reminders and communication tracking

Features 8: Add and remove departments and programs.

Conclusion and Recommendation

The Lead Management System provides an effective solution for managing admission inquiries and scholarship applications at Kathmandu University. The system introduces a centralized digital platform that helps administrators track student leads efficiently and improve communication between departments.

The project demonstrates how web technologies can simplify administrative processes and enhance admission management efficiency. With further development and institutional support, the Lead Management System could serve as a valuable tool for improving admission management, enhancing transparency in scholarship evaluation, and supporting the overall digital transformation of administrative processes at Kathmandu University.

Limitations

i. Limited testing with a small number of users.
ii. The system is currently designed as a web application prototype.
iii. Time constraints limit the implementation of advanced features.
iv. No integration of regression models for lead prediction.

Future Enhancement

i. Integration with email and SMS notification systems.
ii. Development of a mobile application version.
iii. Integration with university admission systems.
iv. Integration with Payment Gateways
v. Cloud Deployment and Scalability
vi. Artificial Intelligence for Lead Prediction

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Appendix A

Screenshots of the Developed System:

Figure A1: *Home screen of the Developed Project*

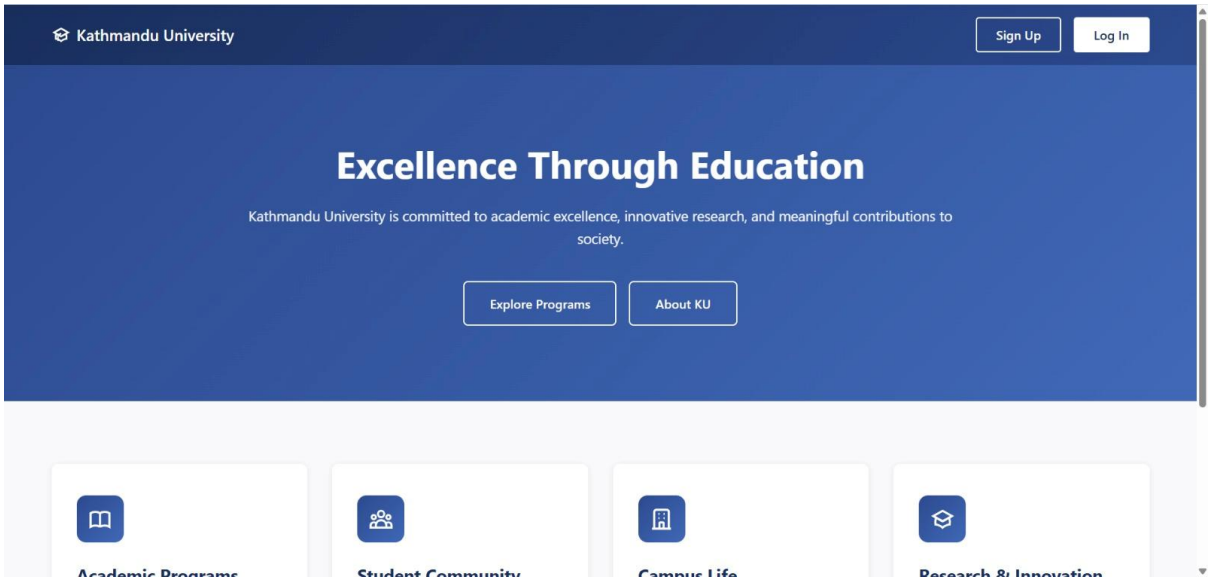


Figure A2: *Login screen of the Developed Project*

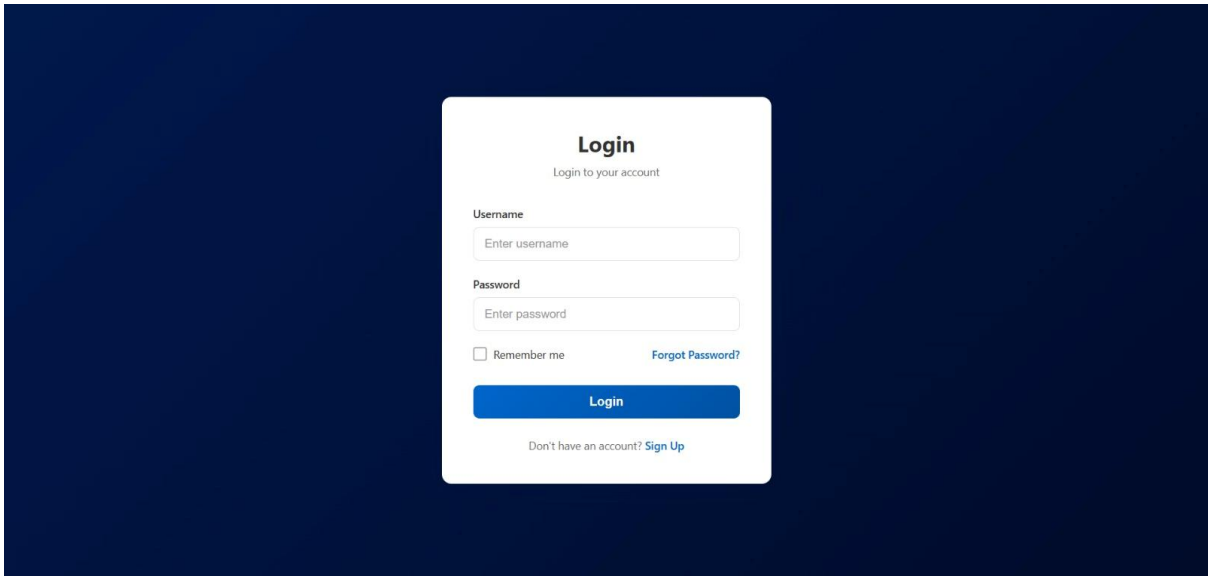


Figure A3: *Sign up Screen of the Developed Project*

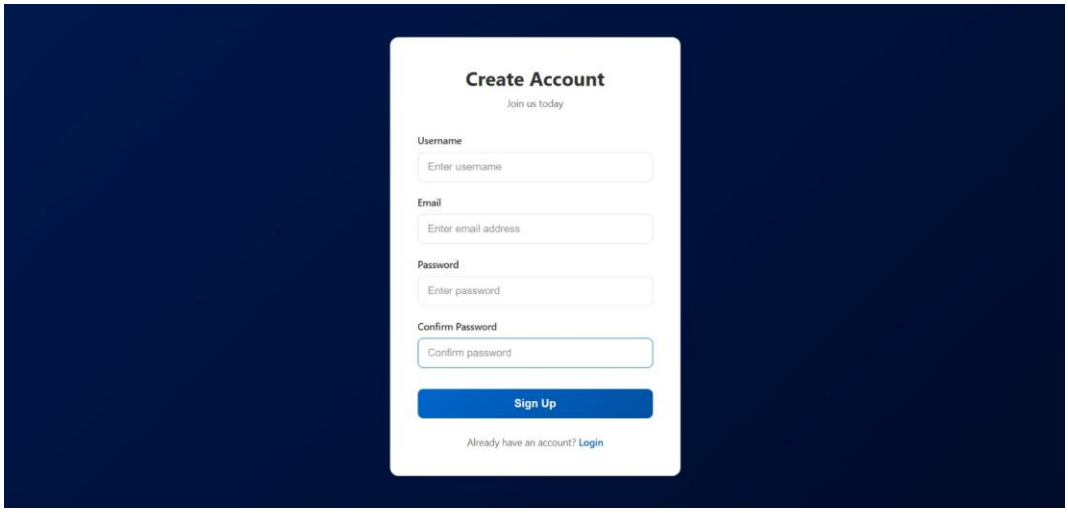


Figure A4: *Admin Dashboard screen of the Developed Project*

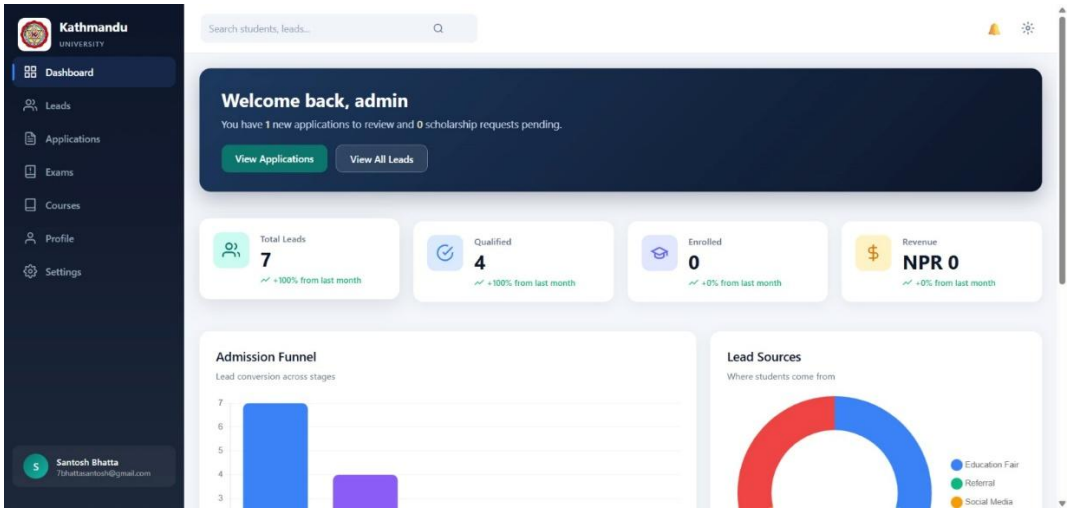


Figure A5: *Lead screen of the Developed Project*

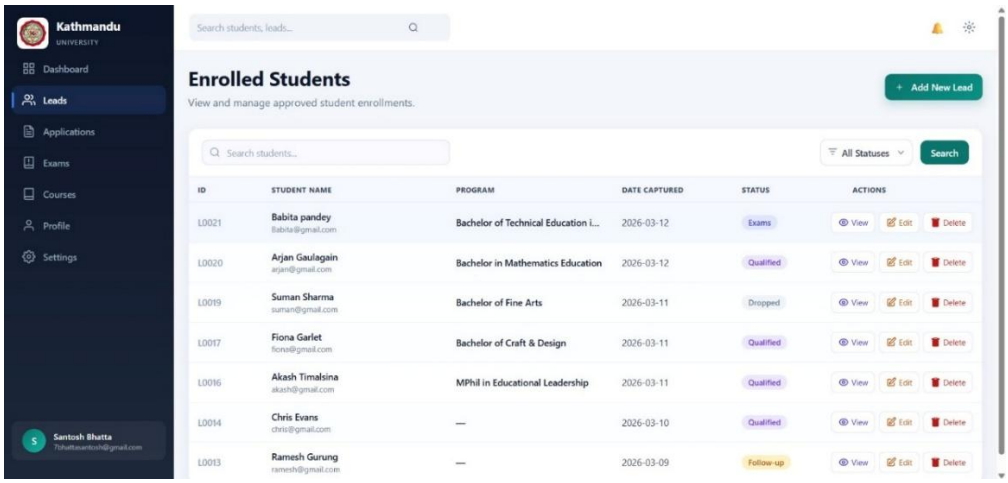


Figure A6: *Application screen of the Developed Project*

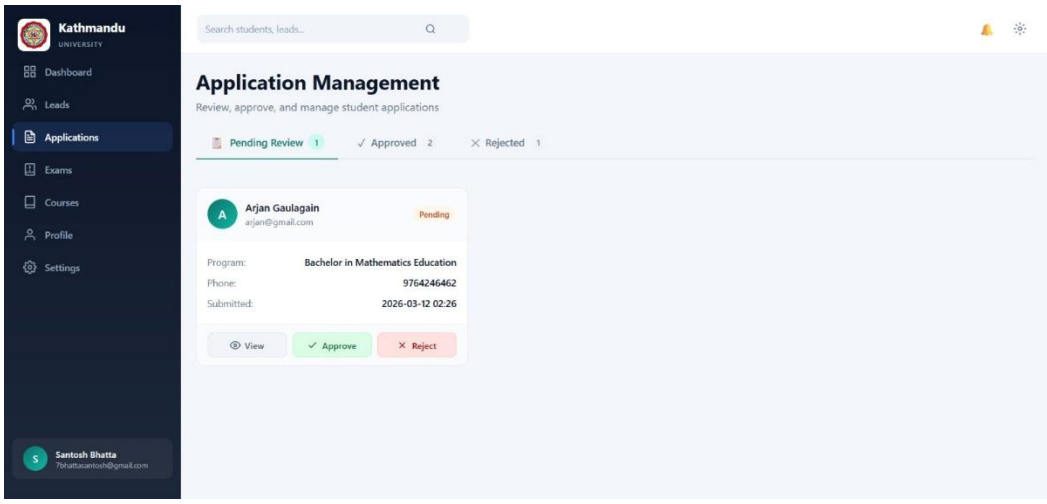


Figure A7: *Exam screen of the Developed Project*

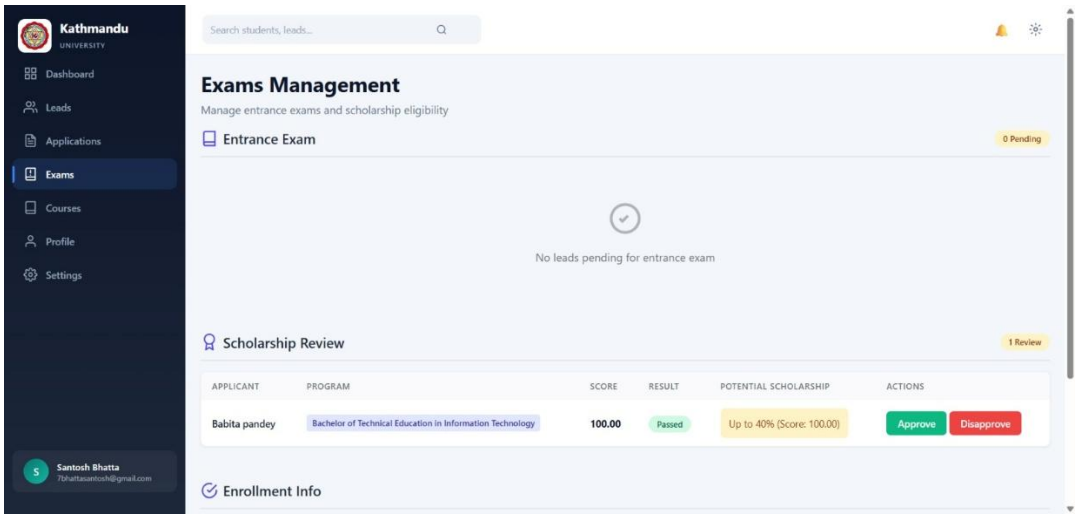


Figure A8: *Course screen of the Developed Project*

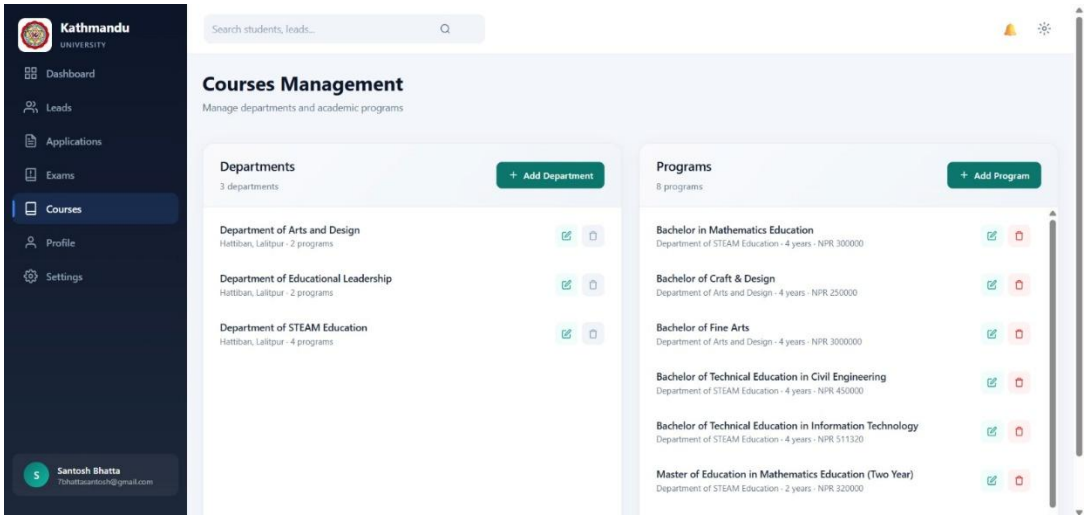


Figure A9: *Profile setting screen of the Developed Project*

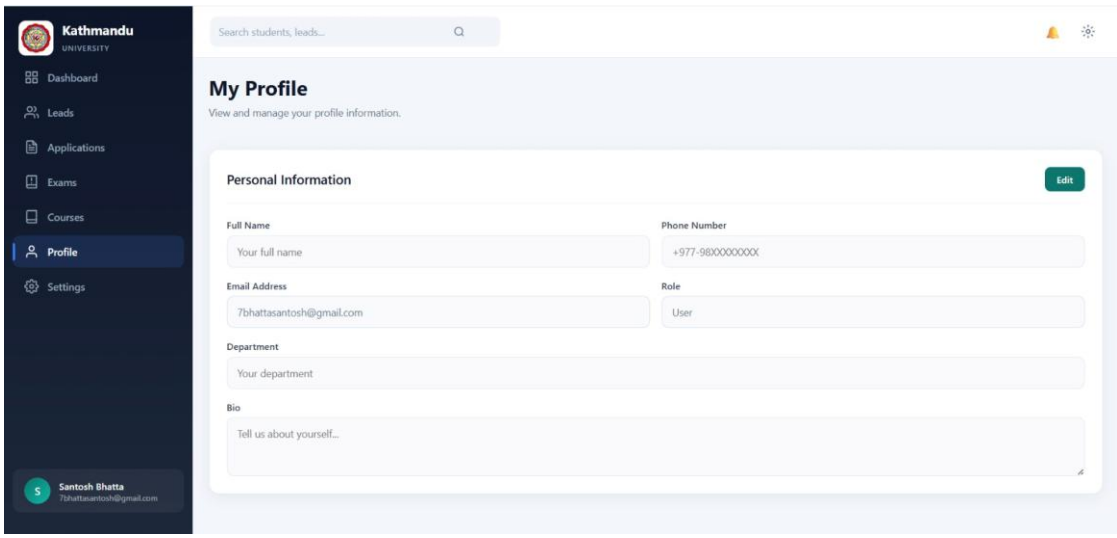
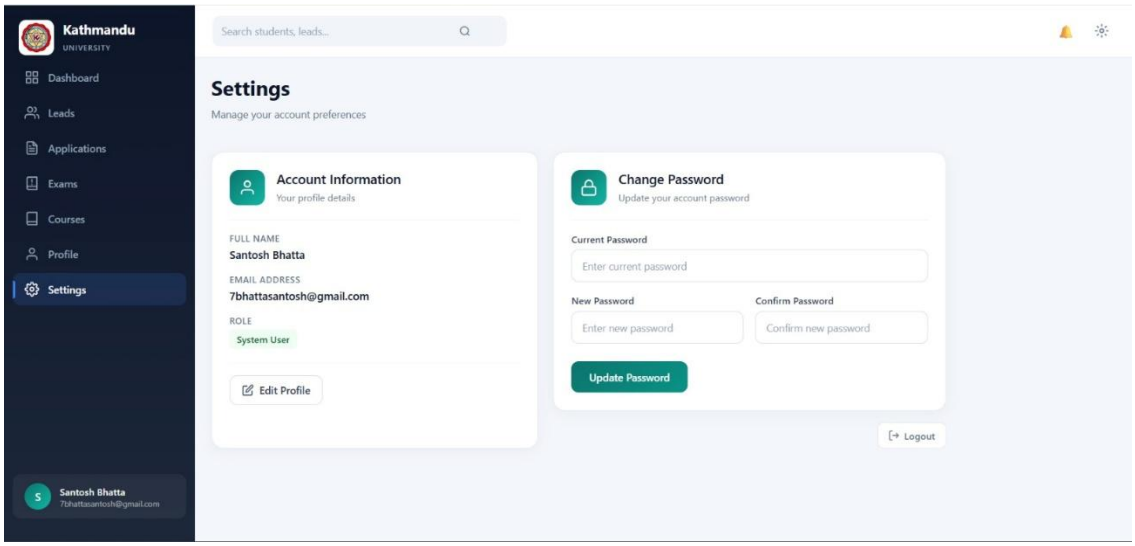


Figure A10: *Setting screen of the Developed Project*



Appendix B

https://github.com/Dipshikha-Ghimire/Lead-Management-system/tree/Django

https://github.com/Dipshikha-Ghimire/Lead-Management-system